

# Digital Actors and Deepfakes: How Artificial Intelligence Is Changing Cinematic Performance

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**Abstract**—Deepfakes, de-aging, and hyper-realistic digital actors are some of the ways artificial intelligence is changing cinematic performance. While these methods increase the potential for creativity, they also make audience trust and consent more difficult to maintain, and they complicate authenticity. Building on previous research on digital resurrection and screen deepfakes (Bode, 2021; Lees, 2023; Lees et al., 2021), as well as surveys of creation and detection techniques (Mirsky Lee, 2020; Khan et al., 2023; Wang et al., 2025), this study combines a small audience experiment with a literature and case review. Participants report their ethical and emotional responses while determining whether brief stills or clips are artificial intelligence AI-generated or real. Initial findings from related research point to substantial misclassification and conflicting emotions, including both fascination and unease as well as worries about identity and consent (Murphy et al., 2023; Mustak et al., 2022; Wagner Blewer, 2019; Lundberg, 2024). We contend that AI can improve visual storytelling while obfuscating the distinction between real and synthetic, necessitating updated creative and legal frameworks as well as more transparent disclosure procedures. A small, offline educational website that showcases recognition tasks, carefully chosen screen cases, and open labeling procedures is one of the project's contributions.

**Index Terms**—Deepfake; digital actors; cinema; audience perception; authenticity; ethics; digital doubles

## I. INTRODUCTION

Digital doubles, face synthesis, and de-aging are examples of AI-driven techniques that are now clearly visible on screen. Filmmakers can stage scenes that were previously impractical, recreate likenesses for continuity, and increase an actor's range across time and age. The potential and challenges of these practices have been documented by scholars, including the "digital face" and its effects on viewers (Bode, 2021), the use of digital media in documentaries that run the risk of conflating representation and reality (Lees, 2023), and dramatic instances of "digital resurrection" that raise issues of identity and consent (Lees et al., 2021). Industry case reports emphasize ethical and experiential trade-offs (Murphy et al., 2023; Mustak et al., 2022; Wagner Blewer, 2019; Lundberg, 2024; Raemy, 2025; Thomas, 2024), while parallel surveys systematize the detection landscape and the creation pipelines (Mirsky Lee, 2020; Khan et al., 2023; Wang et al., 2025). This essay explores how these methods alter performance and

audience perception, asking what expressive power is gained and what authenticity and trust are jeopardized. We relate technological affordances to audience reactions and to new standards for disclosure and consent, using *Rogue One* and *The Irishman* as classic examples and *Sora/MetaHuman* as contemporary tool exemplars.

## II. RESEARCH PROBLEM OBJECTIVE

### A. Problem

With creative advantages but enduring ethical and perceptual issues, AI-driven digital performance blurs the lines between real and synthetic.

### B. Objective

To examine how digital actors and deepfakes alter performance, affect audience engagement and recognition, and challenge conventional notions of authenticity.

## III. RESEARCH QUESTIONS HYPOTHESIS

RQ1: How are cinematic performances altered by deepfake and digital actor techniques?

RQ2: Is it possible for viewers to consistently discern between artificial intelligence-generated and real faces on screen?

RQ3: What creative and moral dilemmas come up during production and reception?

RQ4: What effects does AI use have on trust and emotional engagement?

### A. Hypothesis

Artificial intelligence AI-generated performances improve visual storytelling while fusing simulation and reality, resulting in both creative benefits and moral ambiguity as well as shifting audience trust.

## IV. METHODOLOGY

This research employed a mixed-methods design, combining an audience recognition experiment with a qualitative perception survey. The main objective was to evaluate how viewers perceive, classify, and emotionally respond to AI-generated, digitally modified, and real cinematic performances.

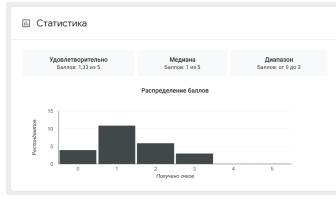


Fig. 1. Enter Caption

### A. Participants

A total of 17 respondents participated in the study. Participants were mostly undergraduate students aged 18–30 from Astana IT University (School of Creative Industries). Most had prior exposure to digital media and film, watching movies or streaming content several times per week. The participants were chosen through convenience sampling, and their responses were collected anonymously through Google Forms between October 26–29, 2025.

## V. MATERIALS AND PROCEDURE

The survey consisted of three main parts:

**Recognition Task — “AI or Reality?”** Participants were presented with stills and short clips from films that incorporated both real and synthetic performances, including *Rogue One: A Star Wars Story* (2016), *The Irishman* (2019), and several visually realistic but non-AI films such as *1917* (2019). Respondents selected one of six possible classifications for each image: Real, AI-generated, De-aged, Digitally aged, Not fully AI, or Unsure.

**Perception and Emotional Response Section** Participants then reflected on their emotional reactions to discovering that a scene was synthetic. They rated how disturbing, fascinating, or neutral such experiences felt, and how much they trusted AI-created acting compared to human performances.

**Ethical Attitudes and Future Outlook** The final section asked participants whether AI-generated acting should always be clearly labeled, and what they viewed as the greatest risk of synthetic performances — loss of authenticity, lack of consent, identity misuse, or job displacement of human actors.

All responses were collected anonymously, with informed consent obtained digitally at the start of the survey. Data were analyzed through descriptive statistics (percentage distribution) and thematic interpretation of open-ended answers.

## VI. RESULTS AND ANALYSIS

### Recognition Accuracy

None of the 17 participants correctly identified all AI-related scenes. Most respondents confused real film frames with synthetic ones and often assumed that *The Irishman* was fully AI-generated rather than de-aged through post-production. These results confirm that even trained viewers find it difficult to distinguish between authentic cinematography and AI-assisted imagery.

### Emotional Reactions

Across answers, most participants reported mixed feelings — a combination of fascination and unease. Many described

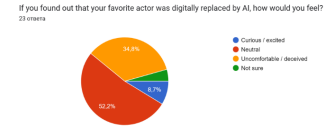


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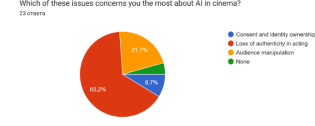


Fig. 3. Enter Caption

the visuals as impressive but slightly disturbing. Only a few respondents expressed complete comfort with AI-generated performances. When asked to imagine a future where AI and real actors become visually indistinguishable, nearly everyone selected the options expressing discomfort or ethical concern.

### Ethical and Creative Attitudes

Almost all participants agreed that AI-generated acting should be clearly labeled in films. The most frequently mentioned concerns included:

loss of authenticity in human performance, absence of actor consent or control over digital likeness, and the risk of misleading audiences.

Despite these worries, several participants acknowledged that AI tools could serve creative purposes when used transparently — for example, for de-aging or special-effects continuity — but rejected the idea of using AI to recreate deceased performers.

## VII.

## VIII. PRELIMINARY OBSERVATIONS

Advanced synthetic faces or brief clips are frequently misclassified by viewers (Mirsky Lee, 2020; Khan et al., 2023; Wang et al., 2025; Murphy et al., 2023). Fascination and uneasiness are mixed emotions, particularly when a clip’s AI origin is revealed (Mustak et al., 2022; Wagner Blewer, 2019; Lundberg, 2024). Consent, digital resurrection, and ownership of identity/likeness are the main topics of discussion (Lees, 2023; Lees et al., 2021; Raemy, 2025; Thomas, 2024). While actors and audiences are concerned about the loss of authenticity and credit, practitioners present AI as a creative tool.

## IX. EXPECTED OUTCOMES CONTRIBUTIONS

An empirical snapshot of attitudes and audience recognition in a small, well-defined cohort. Pedagogical artifact: an offline website (HTML/CSS/JS) that combines screen cases and recognition exercises with transparent AI-synthetic labeling, suitable for self-study or classroom use. Recommendations

for consent and disclosure procedures in AI-mediated performances (where to communicate, what to mark, and how to prevent deceiving audiences).

## X. DISCUSSION

The findings demonstrate that viewers have difficulty recognizing AI manipulation and that their reactions are ambivalent — combining aesthetic admiration with ethical discomfort. This outcome supports earlier studies (Murphy et al., 2023; Lees, 2023) showing that digital faces and deepfakes challenge the emotional contract between actor and spectator.

Participants' strong preference for transparency indicates that the problem lies not in the technology itself, but in hidden manipulation and lack of disclosure. Even real high-quality cinematography was sometimes mistaken for AI, suggesting that cinematic hyperrealism has already reached the boundary where visual trust begins to erode.

## XI. CONCLUSION

AI technologies expand creative possibilities in cinema but blur long-standing boundaries of authenticity and performance. In this small-scale audience study (N = 17):

none of the participants recognized all AI-modified scenes correctly; most expressed curiosity mixed with ethical unease; nearly all supported clear labeling and consent standards.

The study underscores the need for transparent disclosure, negotiated consent for digital likeness, and media-literacy education to help audiences interpret synthetic images critically.

## XII. FUTURE WORK AND LIMITATIONS

The main limitation is the small and homogeneous sample, restricted to students from one university and based on short clips and stills. Future research should include a larger, cross-cultural audience and employ interactive or biometric measures (e.g., reaction times, eye-tracking) to examine subconscious responses. Longitudinal studies could explore how repeated exposure to synthetic media affects perception and emotional trust over time.

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